Foundations of Machine Learning - Final Assignment


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01 Project

01.1 Title: Household Waste Classification Using a Neural Network Implemented from Scratch

01.2 Task: Track 2

Requirements for All Projects

- Implement your neural network from scratch using NumPy
- Use the network for regression, classification, or (for Track 3) next-token language modeling.
- Use a real-world dataset with at least moderate complexity.
- Demonstrate training, evaluation, and interpretation of your model's results.
- Provide insight into hyperparameter choices and learning behaviour.

Track 2: Technical Report

- You choose a standard dataset or task.
- You start with a basic neural network (e.g., a single hidden layer) and incrementally improve it.
- You may explore deeper MLPs, convolutional networks, or attention-based layers (implemented manually).
- You perform hyperparameter searches, ablation studies, and diagnostics
- Technical Report (~6 pages max):
  - Problem Description
  - Dataset Overview
  - Baseline Neural Network
  - Iterative Improvements
  - Evaluation (e.g. accuracy plots, comparison)
  - Lessons Learned and Challenges

01.3 Dataset

The project will use the Garbage Dataset, a multi-class image dataset for automated household-waste classification published in 2026. The dataset contains **13,348 images** distributed across ten categories: metal, glass, biological waste, paper, batteries, residual trash, cardboard, shoes, clothes and plastic. The dataset is published under an MIT licence.
The dataset is suitable for the project because it contains several realistic classification challenges. These include unequal class distributions, which could result in bias towards heavier categories (Kunwar 2026, p. 5), as well as varying backgrounds and lighting conditions, differences in object size, visually similar categories, contaminated objects and composite materials.
Having this characteristics, the dataset is sufficiently complex to reveal the limitations of a simple MLP while remaining manageable for experimenting with a neural network from scratch.

In order to match the German disposal-bin categories, we need to summarize existing categories (shoes, clothes, trash | paper, cardboard | plastic | biological | glass).

02 Background

Nassour et al. (2017) summarize the relevance of manual waste separation at home as follows:
"Source separation has a major impact on the effectiveness of waste management systems, as it causes significant changes in the quantity and quality of waste that reaches final disposal, which is the main factor in the generation of the greenhouse gas, methane. This environmental impact can be significantly reduced by the separate collection and recycling/use of organic waste."

The Garbage Dataset (Kunwar, 2026) contains 13,348 images from ten household-waste categories and explicitly includes challenges such as class imbalance, variable backgrounds, and brightness differences. Using this dataset with a self-tuned multilayer perceptron provides a transparent baseline for examining how different features such as network depth, hidden-layer size, or regularisation affect classification performance. A manually implemented MLP is therefore valuable for understanding the underlying optimisation process and for establishing a comparison model before introducing CNN or attention layers.

Sources:

- Nassour, A., Hemidat, S., Lemke, A., Elnaas, A., Nelles, M. (2017). Separation by Manual Sorting at Home: State of the Art in Germany. In: Maletz, R., Dornack, C., Ziyang, L. (eds) Source Separation and Recycling. The Handbook of Environmental Chemistry, vol 63. Springer, Cham. https://doi.org/10.1007/698_2017_26
- Kunwar, S. (2026). The Garbage Dataset (GD): A Multi-Class Image Benchmark for Automated Waste Segregation. arXiv preprint arXiv: 2602.10500v2.

03 Objective

We want to implement an MLP from scratch using NumPy to classify household waste images into German disposal-bin categories. We will begin with a baseline multilayer perceptron consisting of a single hidden layer and a softmax output layer.
The images will be resized to a common resolution, normalised and divided into stratified training, validation and test sets. For the MLP baseline, the images will be flattened into one-dimensional input vectors. For the convolutional extension, the spatial image structure will be retained. Fixed random seeds will be used to improve reproducibility.

The baseline model will be improved incrementally. Planned experiments include changes to hidden-layer width and depth, activation functions, learning rate, mini-batch size, regularisation and strategies for handling class imbalance. As an advanced extension, we plan to implement a small convolutional feature extractor before the MLP classifier in order to preserve local spatial relationships within the images.

The project is guided by the following research questions:

> **01:** How well can a single-hidden-layer MLP classify household waste images when trained on flattened pixel inputs?

> **02:** How do selected architectural and training choices affect classification performance and training stability?

> **03:** To what extent does a small convolutional feature extractor improve performance compared with a purely fully connected MLP?

We may use performance measures like accuracy and macro-averaged recall like Kunwar (2026) in order to compare our small model against state-of-the-art waste classification models (Table 1).

Table 1: Comparative performance of waste classification models.

Model	Time	Accuracy	Recall	F1 Score	CO_2 Emissions		
					Prepare	Develop	Deploy
EN-V2M	7137.81	94.15	0.93	0.93	0.0011	0.1416	0.1444
EN-V2M-256	6169.28	93.61	0.93	0.93	0.0008	0.0765	0.0784
EN-V2M-384	8084.41	94.58	0.94	0.94	0.0011	0.1583	0.1614
EN-V2S	6057.87	95.13	0.95	0.95	0.0011	0.1180	0.1204
EN-V2S-256	5908.89	93.77	0.94	0.93	0.0015	0.1147	0.1175
EN-V2S-384	4044.54	94.50	0.94	0.94	0.0010	0.0785	0.0807
MN	2420.15	66.94	0.65	0.65	0.0010	0.0475	0.0493
MN-256	2480.46	67.88	0.66	0.66	0.0002	0.0149	0.0153
MN-384	2536.73	70.39	0.68	0.68	0.0002	0.0152	0.0156
RN50	5228.93	92.61	0.92	0.92	0.0015	0.1019	0.1045
RN50-256	5830.07	91.91	0.91	0.91	0.0003	0.0349	0.0356
RN50-384	5643.32	92.39	0.92	0.92	0.0012	0.1115	0.1140
RN101	7331.92	92.77	0.93	0.93	0.0007	0.0906	0.0924
RN101-256	5147.28	92.31	0.92	0.92	0.0017	0.1008	0.1040
RN101-384	7445.92	92.64	0.93	0.92	0.0003	0.0455	0.0463

Abbreviations: EN-V2M = EfficientNetV2M, EN-V2S = EfficientNetV2S, MN = MobileNet, RN50 = ResNet50, RN101 = ResNet101.

Table 1